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VTU.REG.NO

B. E. First Semester Examination – February 2024

Physics for CSE Stream

(Common for CSE, CSD, ISE and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Physical constants : Velocity of light, $c = 3 \times 10^8$ ms^{-1}; Planck constant, $h = 6.63 \times 10^{-34}$ Js; Boltzman constant , $k = 1.38 \times 10^{-23}$ J/K; Avagadro's number, $N_A = 6.023 \times 10^{26}$ gm/Kmole ; Permittivity of vacuum, $\varepsilon_o = 8.85 \times 10^{-12}$ Fm^{-1}; Mass of electron (m) = 9.11×10^{-3} kg; Acceleration due to gravity , g = 9.8 ms^2, Change of electron(e) = 1.6×10^{-19} C.

Module - I

1. a) Explain construction and working of semiconductor diode laser with a neat diagram. (08 Marks)
   b) Describe the terms induced absorption, spontaneous emission and stimulated emission with neat energy level diagram. (09 Marks)
   c) A medium in thermal equilibrium at temperature 300k has two energy levels with a wavelength separation of 1μm. Find ratio of population densities of the upper and lower levels. (03 Marks)

   1.365×10^{-21}

2. a) Explain the different type of optical fibers with neat figures. (09 Marks)
   b) With a neat diagram derive an expression for numerical aperture in an optical fiber. (07 Marks)
   c) The attenuation of light in an optical fiber is 3.6dB/km. What fraction of its initial intensity remains after i) 1km ii) after 3 km. (04 Marks)

   $\frac{P_{out}}{P_{in}} = 0.436$ & 0.0832

Module - II

3. a) Derive one dimensional time- independent Schrodinger wave equation. (03 Marks)
   b) State and explain Heisenberg's uncertainity principle. Give its physical significance. (07 Marks)
   c) An electron is bound in an one dimensional potential well of width 1 Å, but of infinite wall height. Find its energy value in the ground state and in first excited states. (05 Marks)

   37.64 eV 150.54 eV

4. a) Illustrate the non-existence of electrons in the nucleus on the basis of beat decay observation using Heisenberg's uncertainty principle. (08 Marks)
   b) Assuming the time independent Schradinger wave equation discuss the solution for energy of a particle in one-dimensional infinite potentional well. (08 Marks)
   c) A particle of mass 0.5MeV/c^2 has kinetic energy 100eV.Find its de Broglie wavelength , where C is the velocity of light. (04 Marks)

   $\lambda = \frac{h}{\sqrt{2mE}} = 1.24 \times 10^{-10}$ m

Module - III

5. a) Derive an expression for the internal field for a linear array of atoms in a solid dielectric.. (08 Marks)
   b) Derive Clausius – Mossotti equation. (08 Marks)
   c) The dielectric constant of Helium at 0°C is 1.000074. The density of atoms is 2.7 x 10^{25}/m^3. Calculate the dipole moment induced in each atom when the gas is in an electric field of 3 x 10^{4}V/m. (04 Marks)

   $\alpha_e = \frac{\varepsilon_0(\varepsilon_r - 1)}{N} = 7.27 \times 10^{-37}$ cm

6. a) Explain in detail different polarization mechanisms in dielectrics.
   b) Describe properties of ferroelectric materials.
   c) The dielectric constant of sulphur is 3.4. Assuming a cubic lattice for its structure, calculate the electronic polarizability of sulphur.
   Given : for suphur density = 2.07 gm/cc, atomic weight = 32.07.

Module - IV

7. a) Derive an expression for density of states for solid material of unit volume.
   b) Determine the dielectric constant of capacitors using RC circuit. If the length and breadth of capacitor is 47 mm and 5mm and separation between them is 0.075 mm. Given R = 100kΩ.

Time in seconds	Charging	Discharging
0	0	10.3
10	6.1	4
20	8.6	1.6
30	9.6	0.6
40	10	0.2
50	10.2	0.1
60	10.3	0.0

8. a) Explain Type I and type II superconductor.
   b) Explain Fermi dirac statistics with variation of temperature and energy.
   c) Calculate the Planck constant using LED , if the knee voltage of red and blue LEDS are 1.61 and 2.40 V respectively (Given wavelength of red and blue is 660nm and 430 nm).

Module - V

9. a) Explain character animation of jumping with its parts of jump.
   b) Explain the strides and strips in ability to walk.
   c) Write experimental procedure of semiconductor diode laser experiment and calculate the wavelength of laser beam, if the distance between the grating element and screen is 150 cm and the first order and second order diffraction maxima is 10 cm and 20 cm (Given diffraction grating having 2500 lines per inch (LPI)).

10. a) Write characteristics of normal distribution using Bell curves.
    b) Explain descriptive statistics.
    c) Calculate the Fermi energy of copper wire for the given data below with constant 5×10^{-13}.

Temperature (K)	Voltage (mV)	Current (mA)
303	20.9	47
308	21.2	47
313	21.5	46.9
318	21.9	46.9
323	22.2	46.9
328	22.5	46.9